

Yuang Fan

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Research Interest

My research focuses on advancing foundation models through reinforcement learning, multimodal reasoning, and real-world sensing integration. I am particularly interested in improving generalization, robustness, and abstraction capabilities of large language models when applied to longitudinal behavioral, sensor, and visual data. Broadly, I aim to bridge large-scale AI systems with real-world, noisy, and heterogeneous data environments, enabling foundation models to operate reliably in healthcare and embodied sensing contexts.

Education

2025 Jan – Present	Columbia University , New York, NY Ph.D. in Electrical Engineering Advisors: Prof. Xiaofan Jiang and Prof. Xuhai Xu
2023 Sept – 2024 Dec	Columbia University , New York, NY M.S. in Computer Science GPA: 3.95
2019 Sept – 2023 May	New York University , New York, NY B.A. in Computer Science GPA: 3.97 — Summa Cum Laude

Honors and Awards

2023	Phi Beta Kappa Honor Society, New York University
2022	Best Demo Award Runner-up, ACM SenSys '22
2019 – 2023	Dean's List of Academic Year, New York University

Selected Publications

- [1] **Fan, Yuang**, X. Xu, and X. Jiang, “Poster: Split-and-combine rectification of ultra-wide fisheye images into cubemaps,” in *Proceedings of the 31st Annual International Conference on Mobile Computing and Networking*. New York, NY, USA: Association for Computing Machinery, 2025, 1359–1361, ISBN: 9798400711299. [Online]. Available: <https://doi.org/10.1145/3680207.3765686>.
- [2] J. Nie, **Fan, Yuang**, Z. Xuan, M. Zhao, R. Wan, M. Preindl, and X. Jiang, “Soundtrack: A contactless mobile solution for real-time running metric estimation for treadmill running in the wild,” *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.*, vol. 9, no. 2, Jun. 2025. DOI: 10.1145/3729486. [Online]. Available: <https://doi.org/10.1145/3729486>.
- [3] J. Nie, **Fan, Yuang**, M. Zhao, R. Wan, Z. Xuan, M. Preindl, and X. Jiang, “Multi-modal dataset across exertion levels: Capturing post-exercise speech, breathing, and phonocardiogram,” in *Proceedings of the 23rd ACM Conference on Embedded Networked Sensor Systems*, ser. SenSys '25, UC Irvine Student Center., Irvine, CA, USA: Association for Computing Machinery, 2025, 297–304, ISBN: 9798400714795. DOI: 10.1145/3715014.3722065. [Online]. Available: <https://doi.org/10.1145/3715014.3722065>.
- [4] K. Hou, M. Zhao, L. Xu, **Yuang Fan**, and X. Jiang, *Tdbench: A benchmark for top-down image understanding with reliability analysis of vision-language models*, 2025. arXiv: 2504.03748 [cs.LG]. [Online]. Available: <https://arxiv.org/abs/2504.03748>.

- [5] **Fan, Yuang**, J. Nie, X. Sun, and X. Jiang, “Exploring foundation models in detecting concerning daily functioning in psychotherapeutic context based on images from smart home devices,” in *2024 IEEE International Workshop on Foundation Models for Cyber-Physical Systems & Internet of Things (FMSys)*, 2024, pp. 44–49. DOI: 10.1109/FMSys62467.2024.00012.
- [6] J. Nie, H. V. Shao, **Fan, Yuang**, Q. Shao, H. You, M. Preindl, and X. Jiang, “Llm-based conversational ai therapist for daily functioning screening and psychotherapeutic intervention via everyday smart devices,” *ACM Trans. Comput. Healthcare*, Jan. 2025. DOI: 10.1145/3712299. [Online]. Available: <https://doi.org/10.1145/3712299>.
- [7] J. Nie, S. Xia, Y. Liu, S. Ding, L. Hu, M. Zhao, **Fan, Yuang**, M. Abdel-Aty, M. Preindl, and X. Jiang, “A data-driven and human-centric ev charging recommendation system at city-scale,” in *Proceedings of the 14th ACM International Conference on Future Energy Systems (ACM e-Energy '23)*, 2023. DOI: 10.1145/3575813.3597350.
- [8] J. Nie, M. Zhao, S. Xia, X. Sun, H. Shao, **Fan, Yuang**, M. Preindl, and X. Jiang, “Ai therapist for daily functioning assessment and intervention using smart home devices,” in *Proceedings of the 20th ACM Conference on Embedded Networked Sensor Systems (ACM SenSys '22)*, 2023. DOI: 10.1145/3560905.3568074.
- [9] J. Nie, L. Hu, Y. Liu, **Fan, Yuang**, M. Preindl, and X. Jiang, “Human-centric data-driven optimization and recommendation in ev-interfaced grid at city scale: Poster abstract,” in *Proceedings of the 9th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (ACM BuildSys '22)*, 2022. DOI: 10.1145/3563357.3567752.

Research Experiences

2025 – Present

PhD Researcher

Columbia Intelligent and Connected Systems Lab (ICSL), New York, NY

- **RL-Tuned LLMs for Generalizable Mental Health Prediction** Study cross-domain transfer and robustness under dataset heterogeneity and distribution shift in behavioral health prediction tasks. Investigate the use of large language models as abstraction layers that transform high-dimensional longitudinal sensor logs into semantically structured behavioral representations. Develop reinforcement learning pipelines based on Group Relative Policy Optimization to fine-tune reasoning-oriented LLMs, including Qwen-based models, for depression and anxiety outcome prediction from wearable and behavioral data. Design multi-stage reasoning frameworks in which the model first generates structured behavioral summaries and subsequently predicts clinical scores, optimizing summary quality through reward shaping to mitigate generalization gaps inherent in raw behavioral prediction. Execute large-scale distributed training on multi-GPU Slurm clusters with H100 and H200 GPUs.
- **Latent Diffusion Models for Fisheye Image Rectification** Design and train conditional latent diffusion models to convert ultra-wide fisheye images into cube-map projections. Propose a split-and-combine strategy to overcome data scarcity. Implement multi-branch conditioning mechanisms and explore ControlNet-style guidance for geometric consistency. Train large-scale models using PyTorch and Hugging Face Diffusers with perceptual and reconstruction losses. (**poster at Mobicom'25**)

2023 – 2024

Research Assistant

Columbia Intelligent and Connected Systems Lab (ICSL), New York, NY

- **Acoustic-Based Running Metrics Estimation** Built a smartphone-based ML system to estimate cadence and ground contact time from treadmill running sounds. Collected and curated a real-world gym dataset (≥ 60 users); engineered MFCC and spectrogram-based features. Trained multi-task deep learning models in PyTorch achieving 2.11% MAPE (cadence) and 8.78% MAPE (GCT). Deployed on-device inference via TensorFlow Lite. (**poster presented at MobiCom'24, paper at IMWUT'25**)
- **Vision-Language Models for Psychotherapeutic Monitoring:** Investigated the application of vision-language models and large language models to detect concerning activities from smart-home device images. This research is part of an in-home precautionary monitoring system for psychotherapeutic use and has been **accepted to the FMSys 2024 at CPS-IoT Week**.
- **Mobile Mental Health Screening App Development:** Led the adaptation of a browser-based mental health screening app to mobile platforms using Flutter, Python, and Google APIs. Redesigned the user interface to enhance user experience. Trained new CNN models for facial emotion recognition, improving real-time processing efficiency on mobile devices and increasing accuracy from 60% to 70%.

2022

Research Assistant (Summer)

Columbia Intelligent and Connected Systems Lab (ICSL), New York, NY

- Facilitated app development, user interaction, subject testing, and data analysis of 2 research projects. Developed a front-end Flutter AI therapist smartphone app that takes advantage of the existing smart home devices to precautionarily screen and monitor the day-to-day functioning of individuals and provide appropriate interventions; received an average rating of 4.45/5.0 from 20 test subjects over 40 sessions. Awarded the **Best Demo Runner-up** at the 20th ACM Conference on Embedded Networked Sensor Systems.
- Created a Flutter smartphone app with Google Maps API to advise charging routes for electric vehicles in NYC by interacting with a data-driven deep learning simulator based on traffic, EV charger, and household trip information. The results of this work have been presented at the **ACM e-Energy 2023** and **ACM BuildSys 2022**.

Professional Services and Teaching

Teaching Assistant

Columbia University, New York, NY

- AIoT: Artificial Intelligence for Internet of Things, 2025
- Embedded AI, 2026

Reviewer

- ACM IMWUT, 2025
- ACM Health Journal, 2025